Documentation Technique

Mise en place d'un accès distant sécurisé

OpenVPN + Proxy Squid — Infrastructure NetSecure

HEYNEGEN Dinha — BTS SIO option SISR — INGETIS Paris — 2025-2026

---

1. Présentation du projet

Ce document retrace la mise en place d'un accès distant sécurisé via OpenVPN et d'un proxy Squid transparent dans le cadre du projet NetSecure. L'objectif est de permettre aux utilisateurs nomades de se connecter au réseau interne de façon sécurisée, tout en filtrant leur trafic web via un proxy. Le serveur Ubuntu héberge à la fois OpenVPN et Squid. Les clients sont Windows 10 et Debian 12, placés en LAN segment pour simuler le réseau de l'entreprise. Le routage est assuré par pfSense.

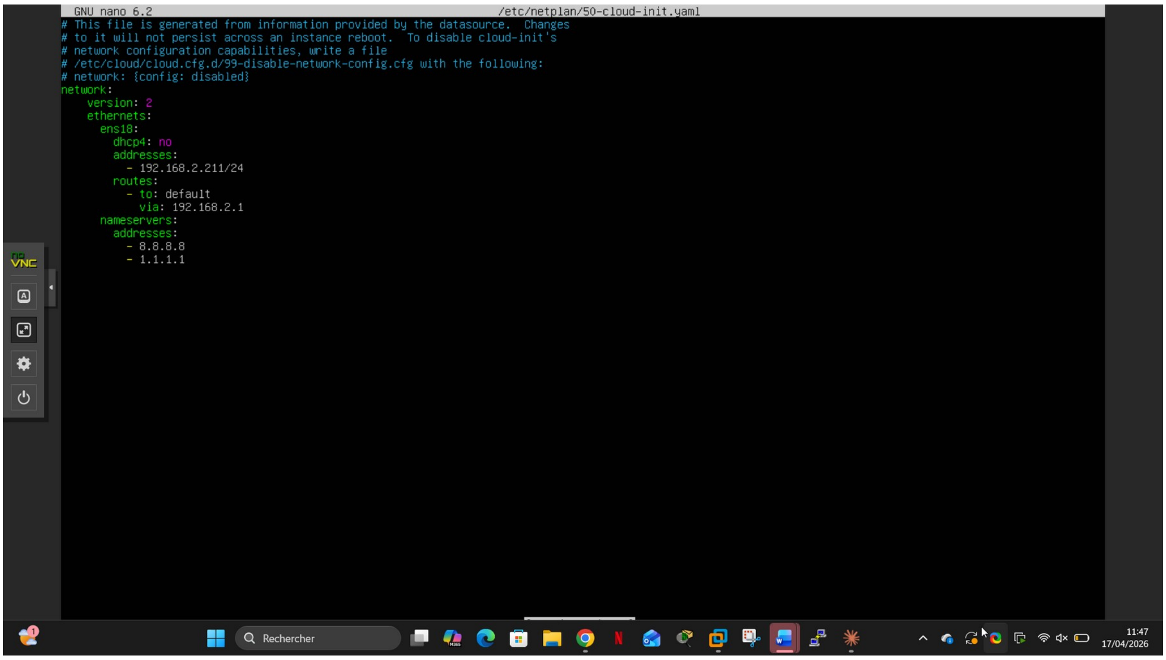
---

2. Configuration réseau — IP statique

> ⚠️ **TOUJOURS fixer les IP des serveurs avant toute chose — en DHCP, les adresses peuvent changer et tout casser !**

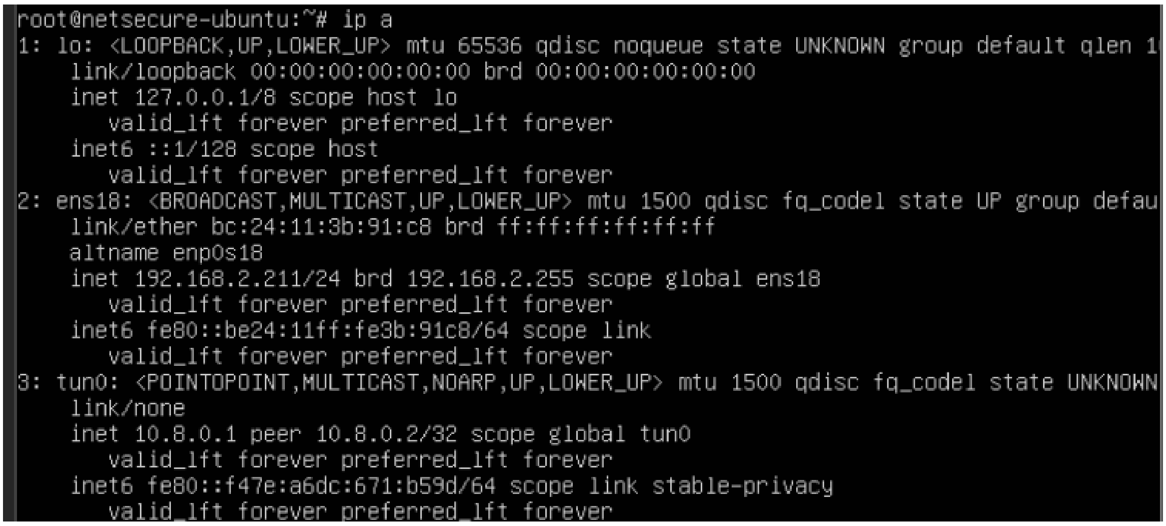
On fixe l'IP statique du serveur Ubuntu en éditant le fichier netplan :

```
nano /etc/netplan/50-cloud-init.yaml
```



Configuration IP statique — fichier netplan 50-cloud-init.yaml

On vérifie les interfaces réseau — ens18 en 192.168.2.211, tun0 (interface VPN) en 10.8.0.1 :



Vérification IP — ip a — interface ens18 en 192.168.2.211, interface tun0 VPN en 10.8.0.1

---

3. Installation et configuration de Squid

```
apt install squid -y
```

```
root@netsecure-ubuntu:~# apt install squid -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
squid est déjà la version la plus récente (5.9-0ubuntu0.22.04.5).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 25 non mis à jour.
root@netsecure-ubuntu:~# systemctl status squid
● squid.service - Squid Web Proxy Server
     Loaded: loaded (/lib/systemd/system/squid.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-04-17 08:35:49 UTC; 3h 21min ago
       Docs: man:squid(8)
   Main PID: 764 (squid)
      Tasks: 4 (limit: 2216)
     Memory: 28.7M
        CPU: 2.748s
     CGroup: /system.slice/squid.service
             ├─764 /usr/sbin/squid --foreground -sYC
             ├─768 "(squid-1)" --kid squid-1 --foreground -sYC
             ├─777 "(logfile-daemon)" /var/log/squid/access.log
             └─782 "(pinger)"

avril 17 08:35:50 netsecure-ubuntu squid[768]: storeLateRelease: released 0 objects
avril 17 09:21:14 netsecure-ubuntu squid[768]: Logfile: opening log stdio:/var/spool/squid/netdb.state
avril 17 09:21:14 netsecure-ubuntu squid[768]: Logfile: closing log stdio:/var/spool/squid/netdb.state
avril 17 09:21:14 netsecure-ubuntu squid[768]: NETDB state saved; 0 entries, 0 msec
avril 17 10:40:00 netsecure-ubuntu squid[768]: Logfile: opening log stdio:/var/spool/squid/netdb.state
avril 17 10:40:00 netsecure-ubuntu squid[768]: Logfile: closing log stdio:/var/spool/squid/netdb.state
avril 17 10:40:00 netsecure-ubuntu squid[768]: NETDB state saved; 0 entries, 0 msec
avril 17 11:51:55 netsecure-ubuntu squid[768]: Logfile: opening log stdio:/var/spool/squid/netdb.state
avril 17 11:51:55 netsecure-ubuntu squid[768]: Logfile: closing log stdio:/var/spool/squid/netdb.state
avril 17 11:51:55 netsecure-ubuntu squid[768]: NETDB state saved; 0 entries, 0 msec
```

Installation et vérification de Squid — apt install squid -y — statut active (running)

On crée une sauvegarde du fichier de configuration avant toute modification :

```
cp /etc/squid/squid.conf /etc/squid/squid.conf.bak
```

```
root@netsecure-ubuntu:~# cp /etc/squid/squid.conf /etc/squid/squid.conf.bak
```

Sauvegarde du fichier de configuration — cp squid.conf squid.conf.bak

On configure Squid via tee — port 3128, ACL réseau local, sites bloqués Facebook et YouTube, refus du reste :

```
root@netsecure-ubuntu:~# tee /etc/squid/squid.conf << 'EOF'
> http_port 3128
> acl localnet src 192.168.2.0/24
> acl sites_bloques dstdomain .facebook.com .youtube.com
> http_access deny sites_bloques
> http_access allow localet
> http_access deny all
> EOF
http_port 3128
acl localnet src 192.168.2.0/24
acl sites_bloques dstdomain .facebook.com .youtube.com
http_access deny sites_bloques
http_access allow localet
http_access deny all
```

Configuration Squid via tee — port 3128, ACL réseau local, sites bloqués (Facebook, YouTube)

On démarre Squid et on vérifie qu'il tourne bien :

```
systemctl start squid && systemctl enable squid
```

```
root@netsecure-ubuntu:~# systemctl start squid
root@netsecure-ubuntu:~# systemctl enable squid
Synchronizing state of squid.service with SysV service script with /lib/systemd/systemd-sy
Executing: /lib/systemd/systemd-sysv-install enable squid
root@netsecure-ubuntu:~# systemctl status squid
● squid.service - Squid Web Proxy Server
     Loaded: loaded (/lib/systemd/system/squid.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-04-17 08:35:49 UTC; 3h 49min ago
       Docs: man:squid(8)
   Main PID: 764 (squid)
      Tasks: 4 (limit: 2216)
     Memory: 28.4M
        CPU: 3.057s
     CGroup: /system.slice/squid.service
             ├─764 /usr/sbin/squid --foreground -sYC
             ├─768 "(squid-1)" --kid squid-1 --foreground -sYC
             ├─777 "(logfile-daemon)" /var/log/squid/access.log
             └─782 "(pinger)"

avril 17 08:35:50 netsecure-ubuntu squid[768]: storeLateRelease: released 0 objects
avril 17 09:21:14 netsecure-ubuntu squid[768]: Logfile: opening log stdio:/var/spool/squid
avril 17 09:21:14 netsecure-ubuntu squid[768]: Logfile: closing log stdio:/var/spool/squid
avril 17 09:21:14 netsecure-ubuntu squid[768]: NETDB state saved; 0 entries, 0 msec
avril 17 10:40:00 netsecure-ubuntu squid[768]: Logfile: opening log stdio:/var/spool/squid
avril 17 10:40:00 netsecure-ubuntu squid[768]: Logfile: closing log stdio:/var/spool/squid
avril 17 10:40:00 netsecure-ubuntu squid[768]: NETDB state saved; 0 entries, 0 msec
avril 17 11:51:55 netsecure-ubuntu squid[768]: Logfile: opening log stdio:/var/spool/squid
avril 17 11:51:55 netsecure-ubuntu squid[768]: Logfile: closing log stdio:/var/spool/squid
avril 17 11:51:55 netsecure-ubuntu squid[768]: NETDB state saved; 0 entries, 0 msec
```

Démarrage et vérification de Squid — systemctl start squid — statut active

---

4. Installation et configuration d'OpenVPN

4.1 Installation

```
apt install openvpn easy-rsa -y
```

```
root@netsecure-ubuntu:~# apt install openvpn easy-rsa -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
easy-rsa est déjà la version la plus récente (3.0.8-1ubuntu1).
openvpn est déjà la version la plus récente (2.5.11-0ubuntu0.22.04.1).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 25 non mis à jour.
```

Installation OpenVPN et easy-rsa — apt install openvpn easy-rsa -y

On vérifie le statut — inactive (dead), le service ne tourne pas encore :

```
root@netsecure-ubuntu:~# systemctl status openvpn@server
○ openvpn@server.service - OpenVPN connection to server
     Loaded: loaded (/lib/systemd/system/openvpn@.service; disabled; vendor preset: enabled)
     Active: inactive (dead)
       Docs: man:openvpn(8)
             https://community.openvpn.net/openvpn/wiki/Openvpn24ManPage
             https://community.openvpn.net/openvpn/wiki/HOWTO
```

Vérification du statut OpenVPN — inactive (dead) — le service ne tourne pas encore

On vérifie que les fichiers de configuration serveur sont bien présents :

```
root@netsecure-ubuntu:~# ls /etc/openvpn/server/
ca.crt  dh.pem  server.conf  server.crt  server.key  ta.key
```

Vérification des fichiers de config serveur dans /etc/openvpn/server/

4.2 Problème : mauvais chemin du fichier server.conf

Tentative de démarrage — erreur : control process exited with error code :

```
root@netsecure-ubuntu:~# systemctl start openvpn@server
Job for openvpn@server.service failed because the control process exited with error code.
See "systemctl status openvpn@server.service" and "journalctl -xeu openvpn@server.service" for details.
```

Tentative de démarrage — erreur : control process exited with error code

On consulte le journal systemd — l'erreur indique qu'OpenVPN cherche /etc/openvpn/server.conf :

```
   The unit openvpn@server.service has entered the 'failed' state with result 'exit-code'.
avril 17 13:06:16 netsecure-ubuntu systemd[1]: Failed to start OpenVPN connection to server.
   Subject: L'unité (unit) openvpn@server.service a échoué
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   L'unité (unit) openvpn@server.service a échoué, avec le résultat failed.
avril 17 13:06:21 netsecure-ubuntu systemd[1]: openvpn@server.service: Scheduled restart job, restart counter is at 47.
   Subject: Le redémarrage automatique d'une unité (unit) a été planifié
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   Le redémarrage automatique de l'unité (unit) openvpn@server.service a été planifié, en
   raison de sa configuration avec le paramètre Restart=.
avril 17 13:06:21 netsecure-ubuntu systemd[1]: Stopped OpenVPN connection to server.
   Subject: L'unité (unit) openvpn@server.service a terminé son arrêt
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   L'unité (unit) openvpn@server.service a terminé son arrêt.
avril 17 13:06:21 netsecure-ubuntu systemd[1]: Starting OpenVPN connection to server...
   Subject: L'unité (unit) openvpn@server.service a commencé à démarrer
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   L'unité (unit) openvpn@server.service a commencé à démarrer.
avril 17 13:06:21 netsecure-ubuntu ovpn-server[4222]: Options error: In [CMD-LINE]:1: Error opening configuration file: /etc/openvpn/server.conf
avril 17 13:06:21 netsecure-ubuntu ovpn-server[4222]: Use --help for more information.
avril 17 13:06:21 netsecure-ubuntu systemd[1]: openvpn@server.service: Main process exited, code=exited, status=1/FAILURE
   Subject: Unit process exited
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   An ExecStart= process belonging to unit openvpn@server.service has exited.

   The process' exit code is 'exited' and its exit status is 1.
avril 17 13:06:21 netsecure-ubuntu systemd[1]: openvpn@server.service: Failed with result 'exit-code'.
   Subject: Unit failed
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   The unit openvpn@server.service has entered the 'failed' state with result 'exit-code'.
avril 17 13:06:21 netsecure-ubuntu systemd[1]: Failed to start OpenVPN connection to server.
   Subject: L'unité (unit) openvpn@server.service a échoué
   Defined-By: systemd
   Support: http://www.ubuntu.com/support

   L'unité (unit) openvpn@server.service a échoué, avec le résultat failed.
lines 1907-1955/1955 (END)
```

Journal systemd — erreur : Error opening configuration file /etc/openvpn/server.conf

```
Options error: In [CMD-LINE]:1: Error opening configuration file: /etc/openvpn/server.conf
```

Erreur précise : Options error — fichier introuvable au chemin attendu

On liste /etc/openvpn/ — le dossier server/ est là mais pas de server.conf à la racine :

```
root@netsecure-ubuntu:~# ls /etc/openvpn/
client  server  update-resolv-conf
```

ls /etc/openvpn/ — le dossier server/ est présent mais server.conf absent à la racine

On liste /etc/openvpn/server/ — le fichier server.conf est bien dans le sous-dossier :

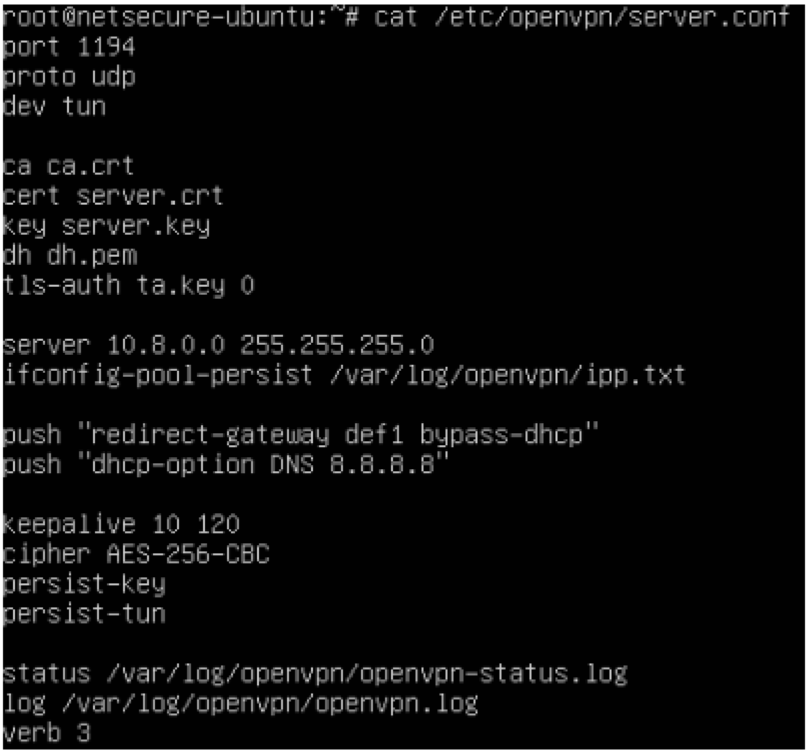
```
root@netsecure-ubuntu:~# ls /etc/openvpn/server
ca.crt  dh.pem  server.conf  server.crt  server.key  ta.key
```

ls /etc/openvpn/server/ — le fichier server.conf est bien là, dans le sous-dossier

On copie le fichier au bon endroit et on retente — erreur persistante. On fait un cat du fichier :

```
root@netsecure-ubuntu:~# cp /etc/openvpn/server/server.conf /etc/openvpn/server.conf
root@netsecure-ubuntu:~# systemctl start openvpn@server
Job for openvpn@server.service failed because the control process exited with error code.
See "systemctl status openvpn@server.service" and "journalctl -xeu openvpn@server.service" for details.
root@netsecure-ubuntu:~# systemctl status openvpn@server
• openvpn@server.service - OpenVPN connection to server
     Loaded: loaded (/lib/systemd/system/openvpn@.service; enabled; vendor preset: enabled)
     Active: activating (auto-restart) (Result: exit-code) since Fri 2026-04-17 13:17:44 UTC; 2s ago
       Docs: man:openvpn(8)
             https://community.openvpn.net/openvpn/wiki/Openvpn24ManPage
             https://community.openvpn.net/openvpn/wiki/HOWTO
    Process: 4671 ExecStart=/usr/sbin/openvpn --daemon ovpn-server --status /run/openvpn/server.status 10 --cd
   Main PID: 4671 (code=exited, status=1/FAILURE)
        CPU: 47ms
lines 1-9/9 (END)
```

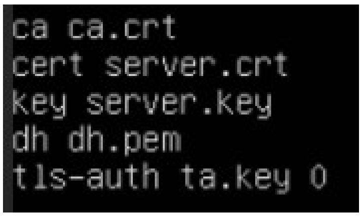
Copie du fichier au bon endroit et nouvelle tentative — erreur persistante



```
root@netsecure-ubuntu:~# cat /etc/openvpn/server.conf
port 1194
proto udp
dev tun

ca ca.crt
cert server.crt
key server.key
dh dh.pem
tls-auth ta.key 0

server 10.8.0.0 255.255.255.0
ifconfig-pool-persist /var/log/openvpn/ipp.txt

push "redirect-gateway def1 bypass-dhcp"
push "dhcp-option DNS 8.8.8.8"

keepalive 10 120
cipher AES-256-CBC
persist-key
persist-tun

status /var/log/openvpn/openvpn-status.log
log /var/log/openvpn/openvpn.log
verb 3
```

cat server.conf — chemins relatifs détectés (sans / devant ca, cert, key, dh, tls-auth)

Les chemins sont relatifs (sans / devant ca, cert, key, dh, tls-auth) — OpenVPN cherche donc dans /etc/openvpn/ au lieu de /etc/openvpn/server/. On corrige avec des chemins absolus :



```
ca ca.crt
cert server.crt
key server.key
dh dh.pem
tls-auth ta.key 0
```

Chemins relatifs dans server.conf — à corriger en chemins absolus

```
root@netsecure-ubuntu:~# tee /etc/openvpn/server.conf << 'EOF'
port 1194
proto udp
dev tun

ca /etc/openvpn/server/ca.crt
cert /etc/openvpn/server/server.crt
key /etc/openvpn/server/server.key
dh /etc/openvpn/server/dh.pem
tls-auth /etc/openvpn/server/ta.key 0

server 10.8.0.0 255.255.255.0
ifconfig-pool-persist /var/log/openvpn/ipp.txt

push "redirect-gateway def1 bypass-dhcp"
push "dhcp-option DNS 8.8.8.8"

keepalive 10 120
cipher AES-256-CBC
persist-key
persist-tun

status /var/log/openvpn/openvpn-status.log
log /var/log/openvpn/openvpn.log
verb 3
EOF
```

Correction via tee — réécriture du server.conf avec chemins absolus vers /etc/openvpn/server/

Nouveau statut après correction — Pre-connection initialization successful :

```
root@netsecure-ubuntu:~# systemctl status openvpn@server
● openvpn@server.service - OpenVPN connection to server
     Loaded: loaded (/lib/systemd/system/openvpn@.service; enabled; vendor preset: enable
     Active: activating (auto-restart) (Result: exit-code) since Fri 2026-04-17 13:34:34
       Docs: man:openvpn(8)
             https://community.openvpn.net/openvpn/wiki/Openvpn24ManPage
             https://community.openvpn.net/openvpn/wiki/HOWTO
    Process: 5823 ExecStart=/usr/sbin/openvpn --daemon ovpn-server --status /run/openvpn/
   Main PID: 5823 (code=exited, status=1/FAILURE)
     Status: "Pre-connection initialization successful"
        CPU: 56ms
```

Nouveau statut après correction — activating (auto-restart) — Pre-connection initialization successful

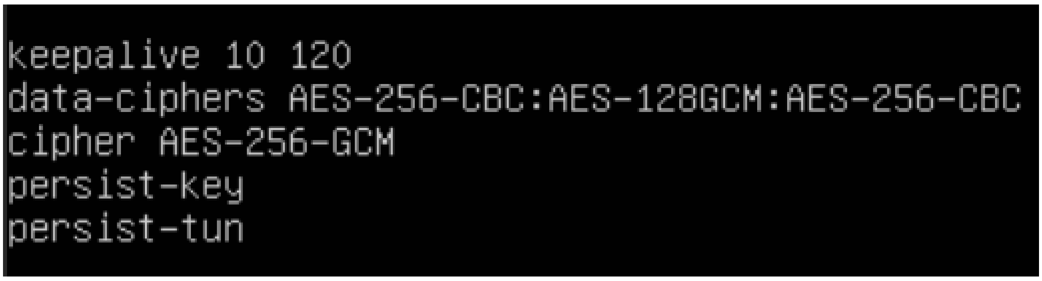
4.3 Problème : cipher incompatible

Une erreur de cipher subsiste. On vérifie la version installée — 2.5.11, version ancienne dans les dépôts Ubuntu :

```
root@netsecure-ubuntu:~# openvpn --version
OpenVPN 2.5.11 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PKTINFO]
library versions: OpenSSL 3.0.2 15 Mar 2022, LZO 2.10
Originally developed by James Yonan
Copyright (C) 2002-2022 OpenVPN Inc <sales@openvpn.net>
Compile time defines: enable_async_push=no enable_comp_stub=no enable_crypto_ofb_cfb=yes enal
le_dlopen=unknown enable_dlopen_self=unknown enable_dlopen_self_static=unknown enable_fast_i
l_lock=yes enable_lz4=yes enable_lzo=yes enable_maintainer_mode=no enable_management=yes enal
ble_pedantic=no enable_pf=yes enable_pkcs11=yes enable_plugin_auth_pam=yes enable_plugin_dow
o enable_shared=yes enable_shared_with_static_runtimes=no enable_silent_rules=no enable_smal
ble_systemd=yes enable_werror=no enable_win32_dll=yes enable_x509_alt_username=yes with_aix_
k=no with_openssl_engine=yes with_sysroot=no
```

Vérification de la version OpenVPN — 2.5.11 — version ancienne dans les dépôts Ubuntu

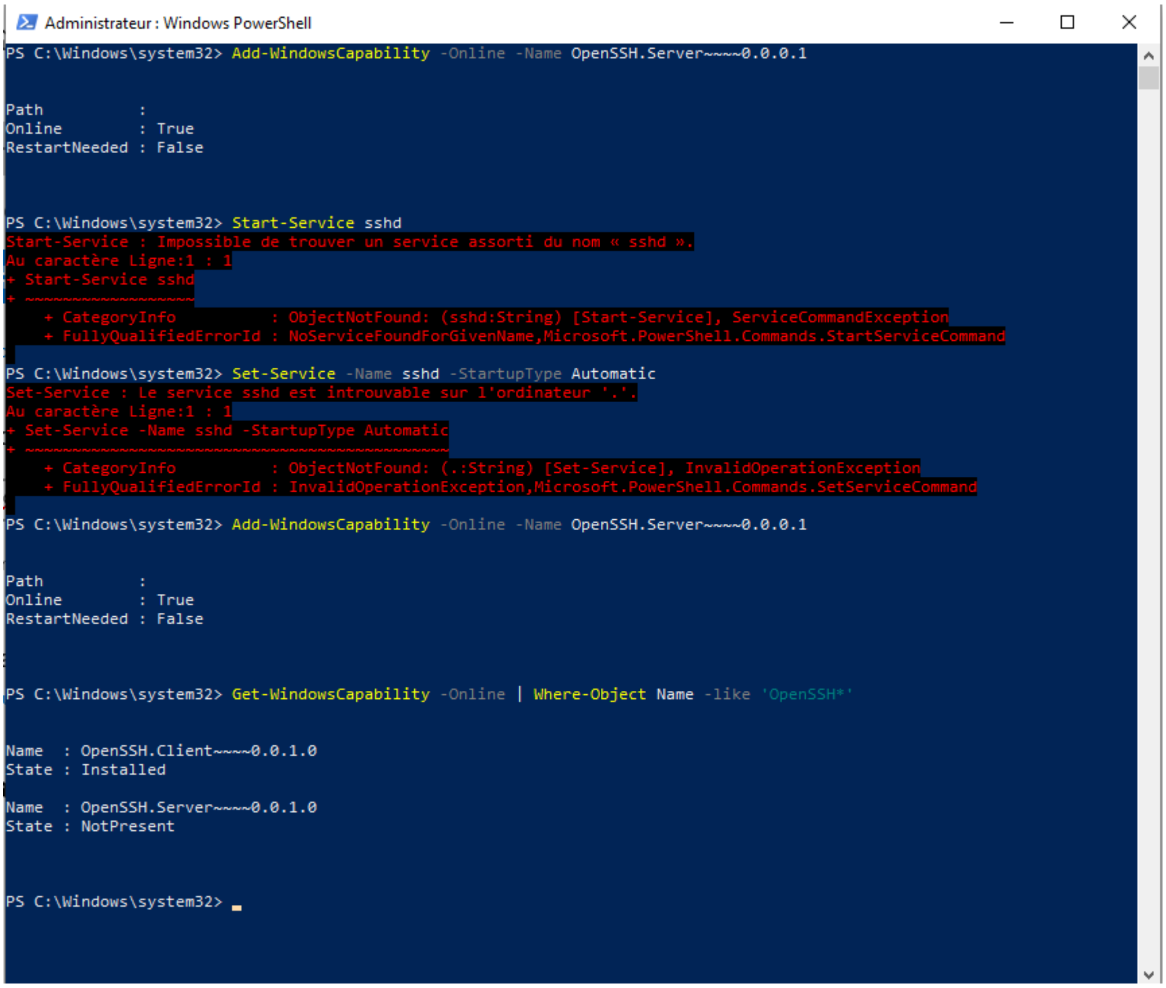
On remplace cipher AES-256-CBC par data-ciphers AES-256-CBC:AES-128GCM:AES-256-CBC dans server.conf :



Correction du cipher — remplacement de AES-256-CBC par data-ciphers AES-256-CBC:AES-128GCM:AES-256-CBC

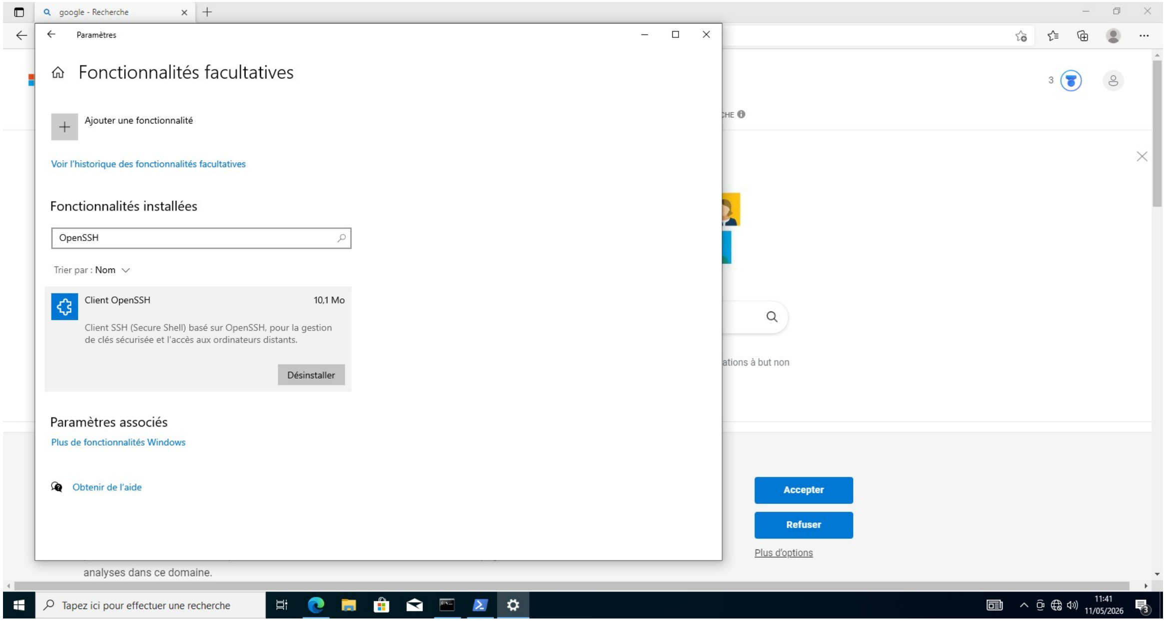
4.4 Problème : distribution des fichiers clients via SCP

Pour transférer les fichiers .ovpn aux clients, on tente d'installer OpenSSH Server sur Windows via PowerShell — bug connu, le service sshd est introuvable :

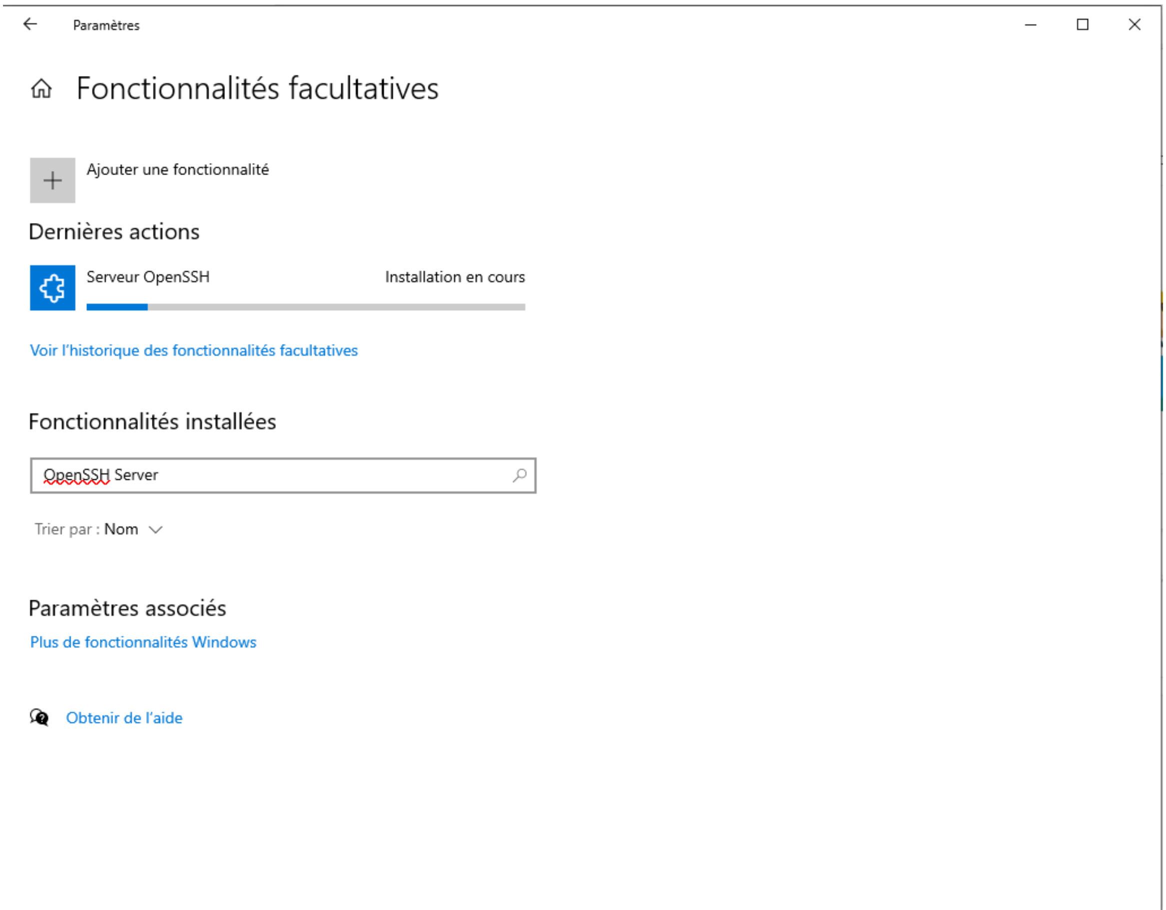


Tentative installation OpenSSH Server via PowerShell — bug connu Windows : service sshd introuvable

On passe par les Fonctionnalités facultatives Windows pour installer OpenSSH Server manuellement :



Fonctionnalités facultatives Windows — installation OpenSSH Server en cours



Fonctionnalités facultatives — OpenSSH Server trouvé

OpenSSH Server installé, démarrage du service :

```
Administrateur : Windows PowerShell
PS C:\Windows\system32> Start-Service sshd
Start-Service : Impossible de trouver un service assorti du nom « sshd ».
Au caractère Ligne:1 : 1
+ Start-Service sshd
+ ~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (sshd:String) [Start-Service], ServiceCommandException
    + FullyQualifiedErrorId : NoServiceFoundForGivenName,Microsoft.PowerShell.Commands.StartServiceCommand

PS C:\Windows\system32> Set-Service -Name sshd -StartupType Automatic
Set-Service : Le service sshd est introuvable sur l'ordinateur '.'.
Au caractère Ligne:1 : 1
+ Set-Service -Name sshd -StartupType Automatic
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : ObjectNotFound: (.:String) [Set-Service], InvalidOperationException
    + FullyQualifiedErrorId : InvalidOperationException,Microsoft.PowerShell.Commands.SetServiceCommand

PS C:\Windows\system32> Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.0.1


Path          :
Online        : True
RestartNeeded : False



PS C:\Windows\system32> Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'


Name  : OpenSSH.Client~~~~0.0.1.0
State : Installed

Name  : OpenSSH.Server~~~~0.0.1.0
State : NotPresent



PS C:\Windows\system32> Get-WindowsCapability -Online | Where-Object Name -like 'OpenSSH*'


Name  : OpenSSH.Client~~~~0.0.1.0
State : Installed

Name  : OpenSSH.Server~~~~0.0.1.0
State : Installed



PS C:\Windows\system32> Start-Service sshd
PS C:\Windows\system32> Set-Service -Name sshd -StartupType Automatic
PS C:\Windows\system32>
```

PowerShell — OpenSSH Server désormais installé, démarrage du service sshd

```
PS C:\Windows\system32> Get-Service sshd

Status   Name               DisplayName
------   ----               -----------
Running  sshd               OpenSSH SSH Server
```

Vérification — Get-Service sshd — Running

Tentative SCP depuis Ubuntu vers Windows — Permission denied :

```
root@netsecure-ubuntu:~# scp /etc/openvpn/client2-final.openvpn user@192.168.2.218:/Users/admin/Desk
top/
user@192.168.2.218's password:
Permission denied, please try again.
user@192.168.2.218's password:
Permission denied, please try again.
user@192.168.2.218's password:
user@192.168.2.218: Permission denied (publickey,password,keyboard-interactive).
lost connection
```

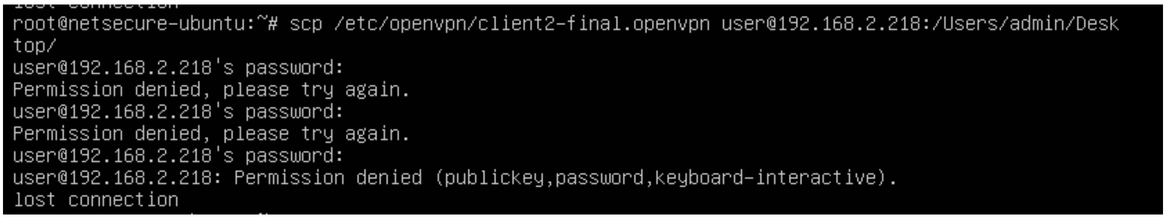
Tentative SCP depuis Ubuntu vers Windows — Permission denied

Modification sshd_config Windows, ajout AuthorizedKeysFile et restart :

```
PS C:\Windows\system32> Add-Content "C:\ProgramData\ssh\sshd_config" "`nMatch Group administrators`n        AuthorizedKey
sFile __PROGRAMDATA__/ssh/administrators_authorized_keys"
PS C:\Windows\system32> Restart-Service sshd
```

Modification sshd_config Windows — ajout AuthorizedKeysFile et Restart-Service sshd

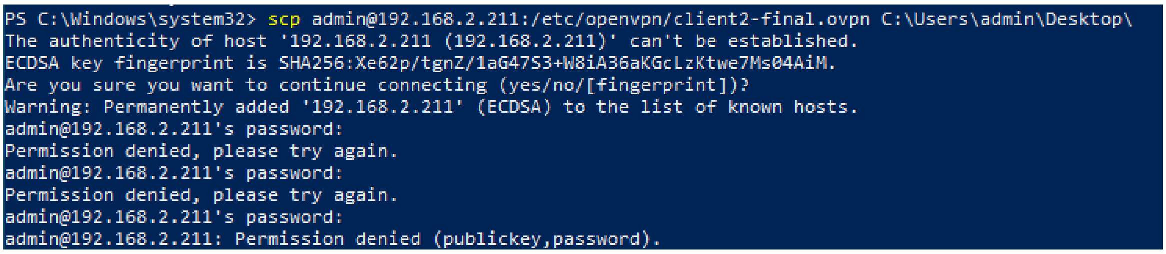
Nouvelle tentative — toujours Permission denied :



```
root@netsecure-ubuntu:~# scp /etc/openvpn/client2-final.openvpn user@192.168.2.218:/Users/admin/Desk
top/
user@192.168.2.218's password:
Permission denied, please try again.
user@192.168.2.218's password:
Permission denied, please try again.
user@192.168.2.218's password:
user@192.168.2.218: Permission denied (publickey,password,keyboard-interactive).
lost connection
```

Nouvelle tentative SCP — toujours Permission denied

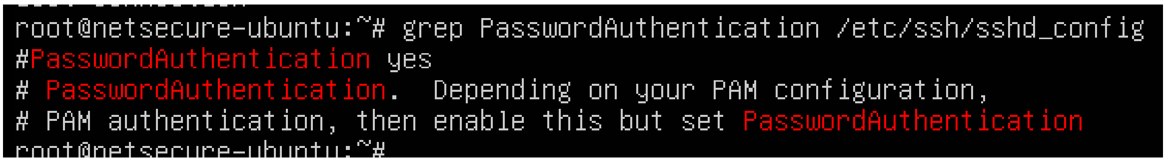
Tentative SCP depuis Windows vers Ubuntu — Permission denied (publickey, password) :



```
PS C:\Windows\system32> scp admin@192.168.2.211:/etc/openvpn/client2-final.ovpn C:\Users\admin\Desktop\
The authenticity of host '192.168.2.211 (192.168.2.211)' can't be established.
ECDSA key fingerprint is SHA256:Xe62p/tgnZ/1aG47S3+W8iA36aKGcLzKtwe7Ms04AiM.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '192.168.2.211' (ECDSA) to the list of known hosts.
admin@192.168.2.211's password:
Permission denied, please try again.
admin@192.168.2.211's password:
Permission denied, please try again.
admin@192.168.2.211's password:
admin@192.168.2.211: Permission denied (publickey,password).
```

Tentative SCP depuis Windows vers Ubuntu — Permission denied (publickey, password)

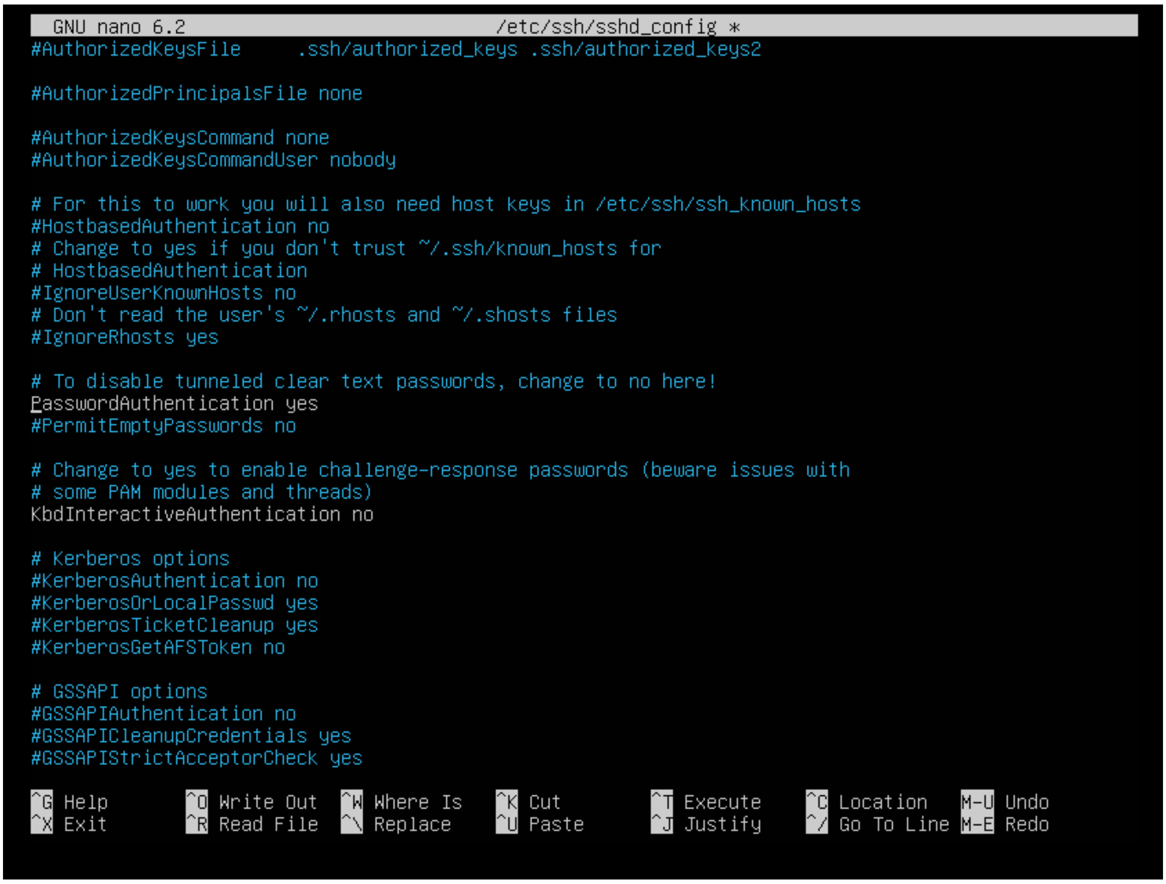
On vérifie PasswordAuthentication dans sshd_config Ubuntu — ligne commentée :



```
root@netsecure-ubuntu:~# grep PasswordAuthentication /etc/ssh/sshd_config
#PasswordAuthentication yes
# PasswordAuthentication.  Depending on your PAM configuration,
# PAM authentication, then enable this but set PasswordAuthentication
root@netsecure-ubuntu:~#
```

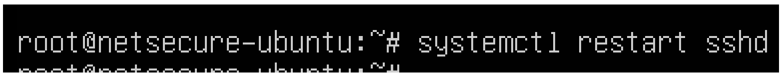
Vérification PasswordAuthentication dans sshd_config Ubuntu — ligne commentée

On décommente PasswordAuthentication yes et on restart sshd :



```
  GNU nano 6.2                           /etc/ssh/sshd_config *
#AuthorizedKeysFile      .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes

^G Help      ^O Write Out  ^W Where Is  ^K Cut      ^T Execute  ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace   ^U Paste    ^J Justify  ^/ Go To Line M-E Redo
```

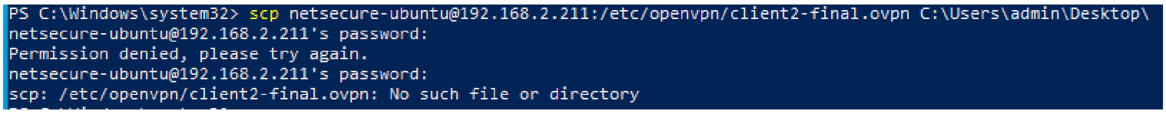
Décommentage de PasswordAuthentication yes dans /etc/ssh/sshd_config Ubuntu



```
root@netsecure-ubuntu:~# systemctl restart sshd
```

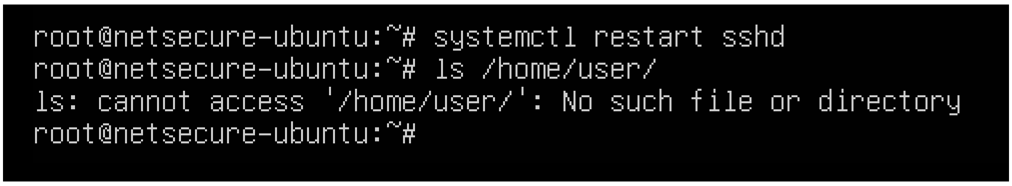
Restart sshd Ubuntu

Nouvelle tentative SCP — No such file or directory, chemin incorrect :



```
PS C:\Windows\system32> scp netsecure-ubuntu@192.168.2.211:/etc/openvpn/client2-final.ovpn C:\Users\admin\Desktop\
netsecure-ubuntu@192.168.2.211's password:
Permission denied, please try again.
netsecure-ubuntu@192.168.2.211's password:
scp: /etc/openvpn/client2-final.ovpn: No such file or directory
```

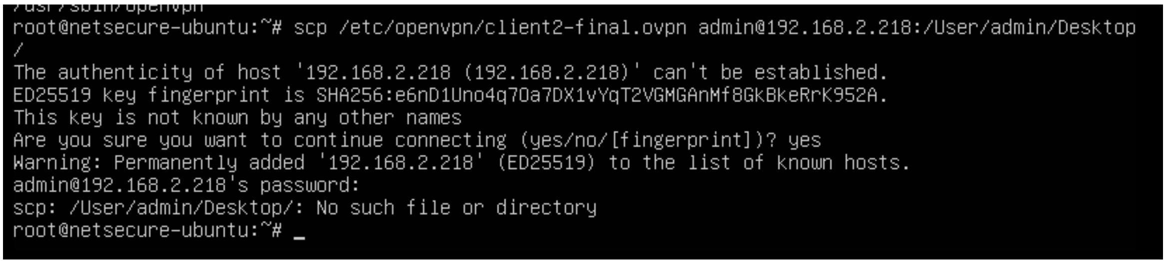
Nouvelle tentative SCP — No such file or directory — chemin incorrect



```
root@netsecure-ubuntu:~# systemctl restart sshd
root@netsecure-ubuntu:~# ls /home/user/
ls: cannot access '/home/user/': No such file or directory
root@netsecure-ubuntu:~#
```

ls /home/user/ — No such file or directory — utilisateur inexistant

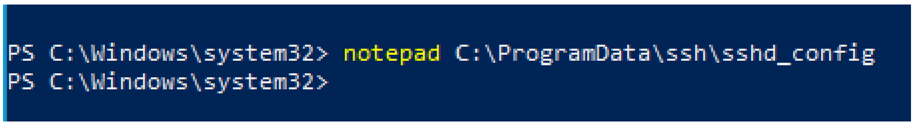
Tentative avec le bon chemin — connexion ok mais dossier introuvable côté Windows :



```
root@netsecure-ubuntu:~# scp /etc/openvpn/client2-final.ovpn admin@192.168.2.218:/User/admin/Desktop
/
The authenticity of host '192.168.2.218 (192.168.2.218)' can't be established.
ED25519 key fingerprint is SHA256:e6nD1Uno4q70a7DX1vYqT2VGMGAnMf8GkBkeRrK952A.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.2.218' (ED25519) to the list of known hosts.
admin@192.168.2.218's password:
scp: /User/admin/Desktop/: No such file or directory
root@netsecure-ubuntu:~# _
```

Tentative SCP avec le bon chemin — connexion ok mais dossier introuvable côté Windows

On ouvre sshd_config Windows via notepad :



```
PS C:\Windows\system32> notepad C:\ProgramData\ssh\sshd_config
PS C:\Windows\system32>
```

Ouverture sshd_config Windows via notepad

PasswordAuthentication commentée dans sshd_config Windows — on décommente :

```
sshd_config - Bloc-notes
Fichier  Edition  Format  Affichage  Aide

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2
# but this is overridden so installations will only check .ssh/authorized_keys
AuthorizedKeysFile	.ssh/authorized_keys

#AuthorizedPrincipalsFile none

# For this to work you will also need host keys in %programData%/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

#AllowAgentForwarding yes

Ln 1, Col 1    100%    Windows (CRLF)    UTF-8
```

sshd_config Windows — PasswordAuthentication commentée

```
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

#AllowAgentForwarding yes
```

Décommentage PasswordAuthentication yes dans sshd_config Windows

Lignes Match Group administrators commentées — on les commente pour désactiver la restriction :

```
#       ForceCommand cvs server

#Match Group administrators
       AuthorizedKeysFile __PROGRAMDATA__/ssh/administrators_authorized_keys


#Match Group administrators
       AuthorizedKeysFile __PROGRAMDATA__/ssh/administrators_authorized_keys
```

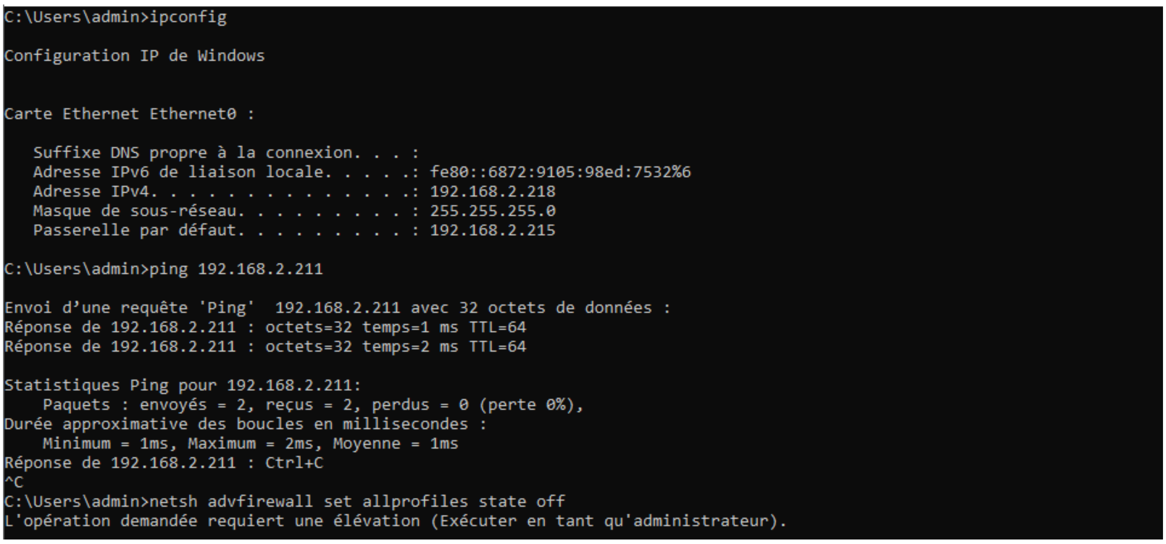
Lignes Match Group administrators commentées dans sshd_config Windows

Restart-Service sshd Windows :

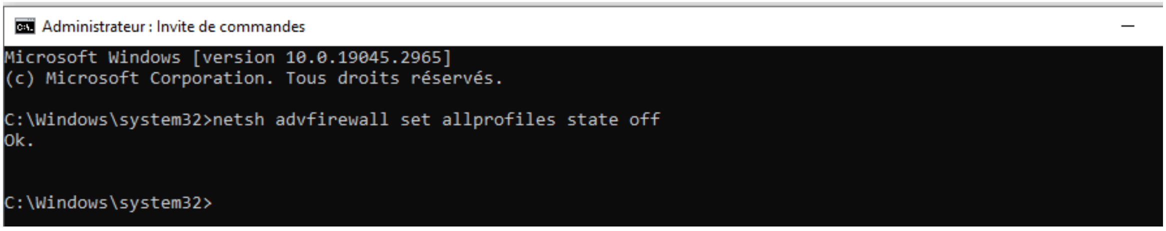
```
PS C:\Windows\system32> Restart-Service sshd
```

Restart-Service sshd Windows

On vérifie la connectivité — ping ok, mais firewall Windows bloque encore. On le désactive en administrateur :



ipconfig Windows — IP 192.168.2.218 — ping serveur Ubuntu ok — firewall bloquait la connexion



Désactivation firewall Windows — netsh advfirewall set allprofiles state off — Ok

Après toutes ces tentatives infructueuses, on abandonne SCP et on opte pour un serveur de partage HTTP Python sur le port 8080 :

---

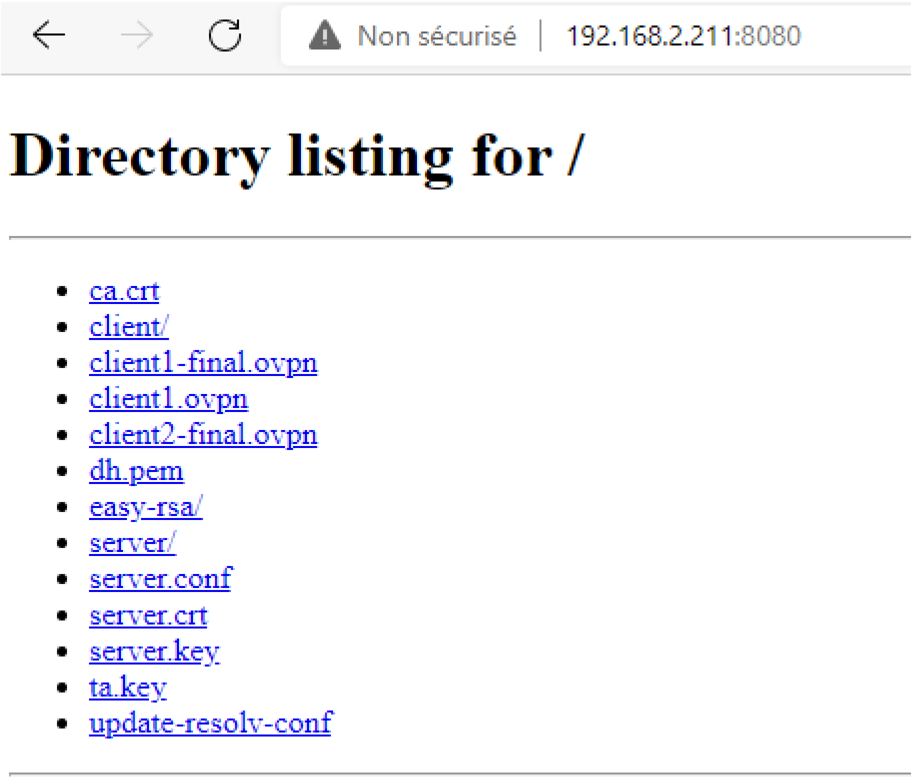
5. Distribution des fichiers via serveur HTTP

On ouvre un serveur de partage Python dans le dossier /etc/openvpn et on autorise le port 8080 :

```
cd /etc/openvpn && python3 -m http.server 8080
ufw allow 8080
```

Serveur de partage HTTP Python sur port 8080 — ufw allow 8080 — Serving HTTP on 0.0.0.0

Les clients accèdent au serveur via l'IP et le port 8080 et voient la liste des fichiers .ovpn disponibles :

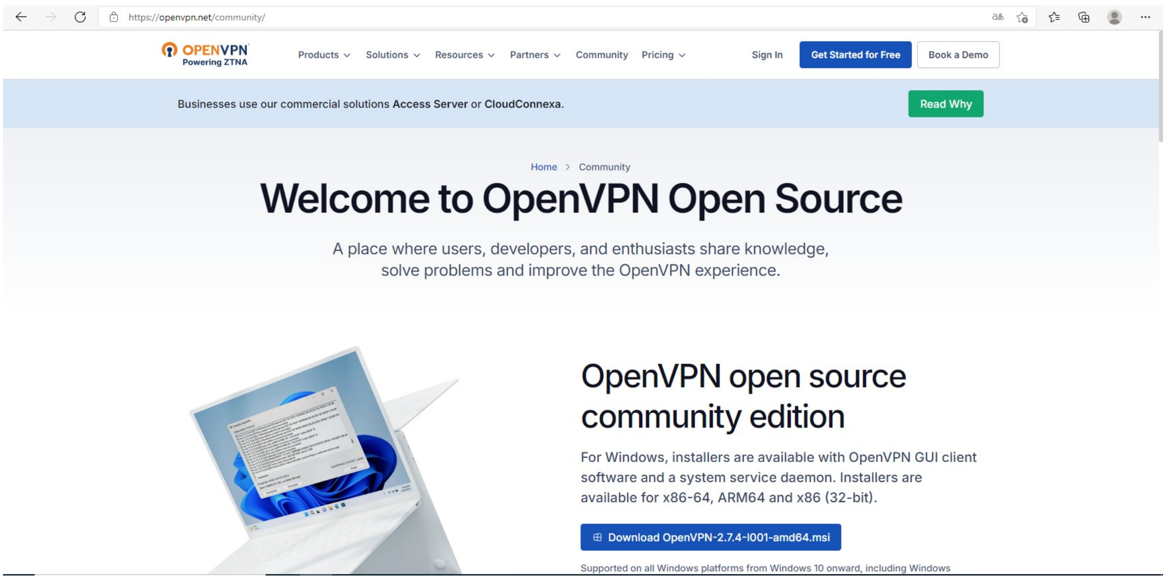


Interface web du serveur de partage — liste des fichiers .ovpn disponibles depuis le navigateur

---

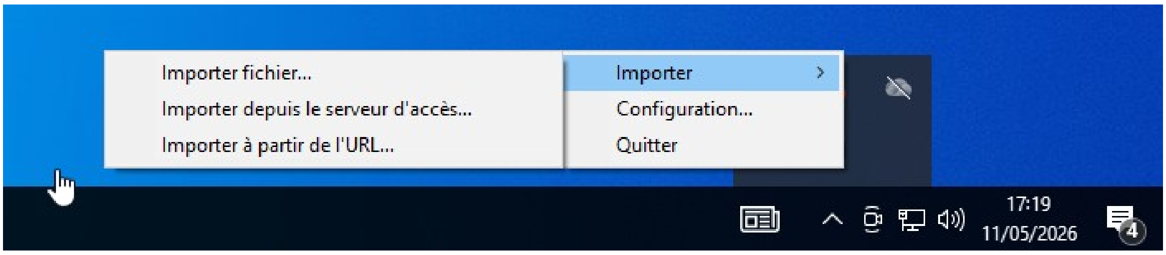
6. Connexion VPN — client Windows

On télécharge le client OpenVPN depuis openvpn.net/community :

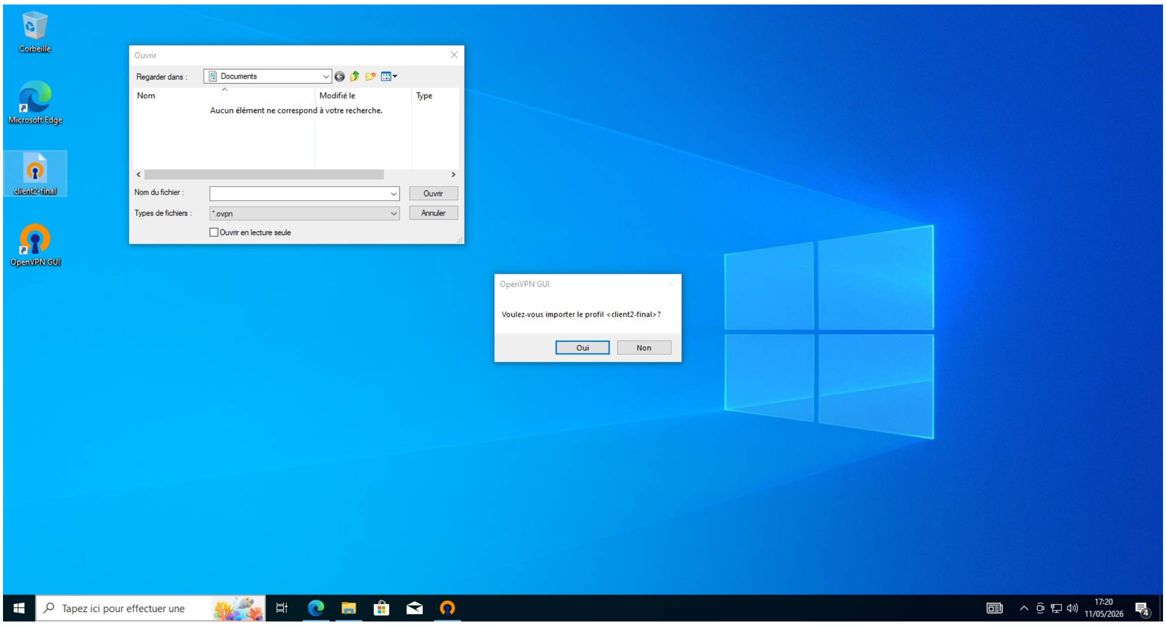


Téléchargement du client OpenVPN depuis openvpn.net/community — version 2.7.4

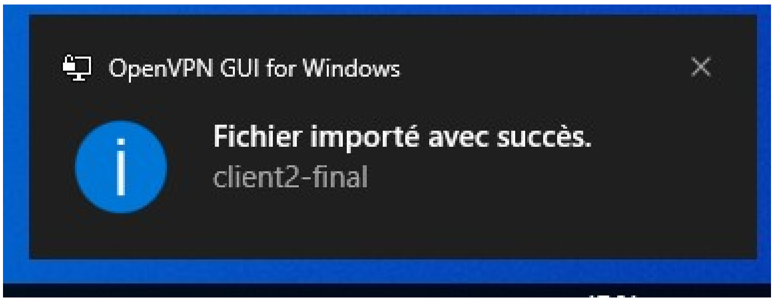
On importe le profil client2-final.ovpn depuis la barre des tâches — Importer fichier :



Import du profil .ovpn — menu barre des tâches OpenVPN GUI — Importer fichier

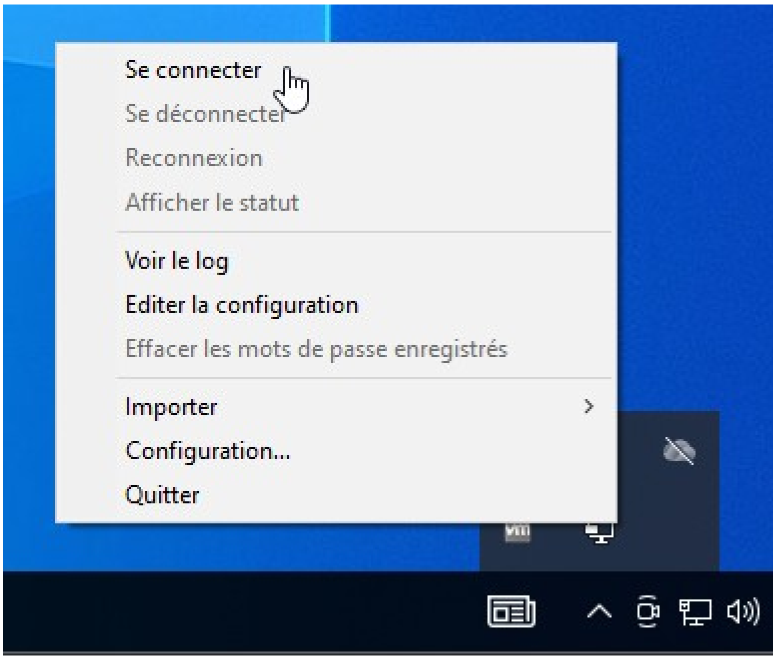


Dialogue import profil — Voulez-vous importer le profil client2-final ?



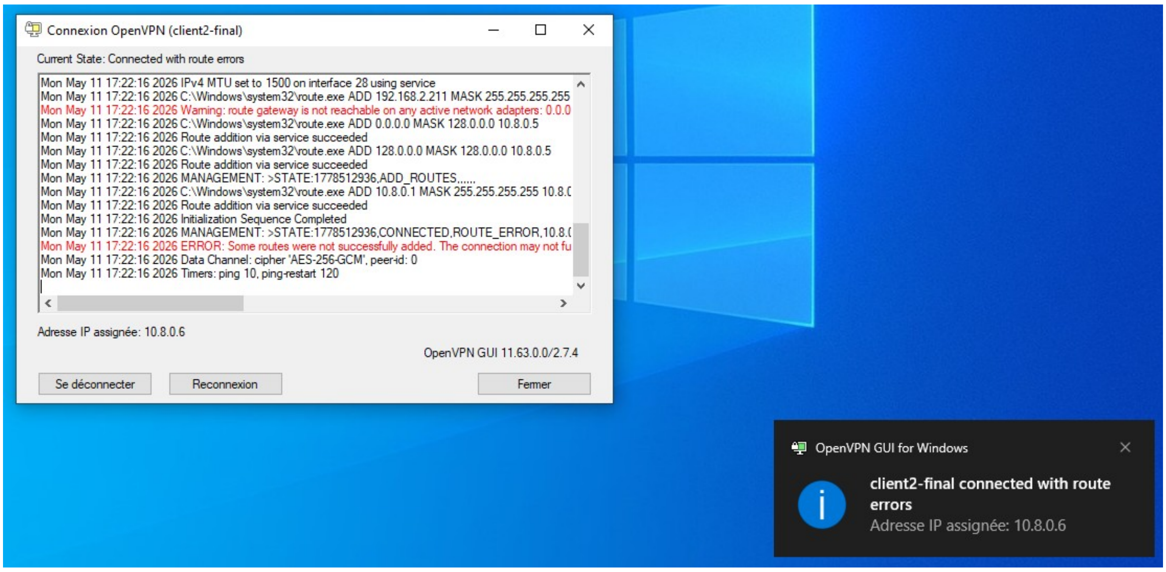
Confirmation — Fichier importé avec succès — client2-final

On se connecte depuis la barre des tâches :



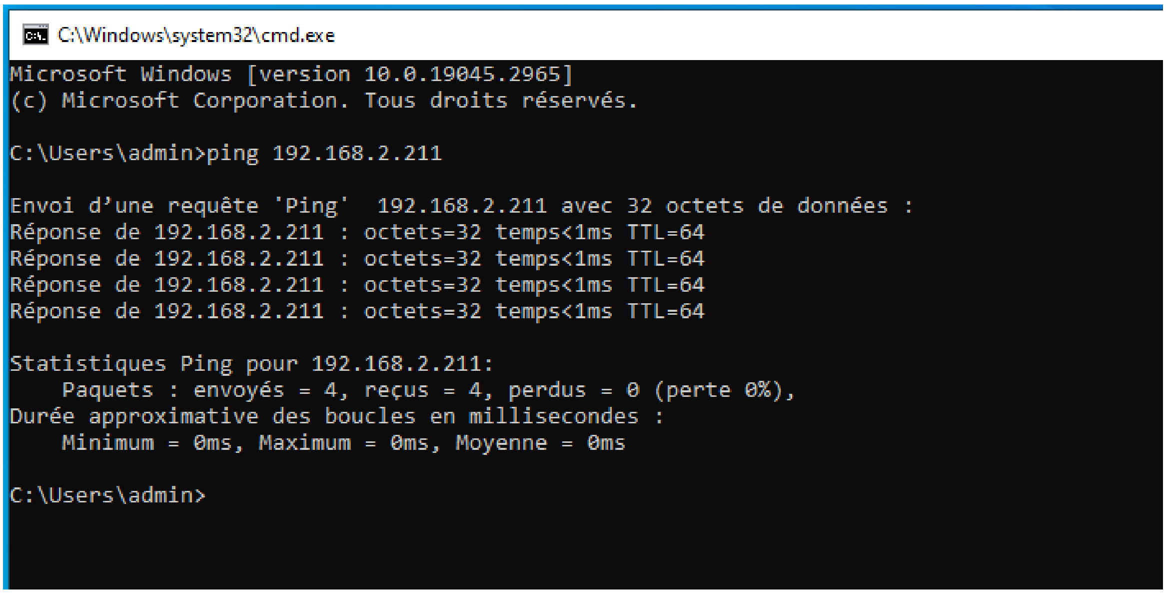
Menu barre des tâches — Se connecter

La connexion s'établit — adresse IP assignée : 10.8.0.6 — Connected with route errors (non bloquant) :



Connexion VPN établie — Adresse IP assignée : 10.8.0.6 — Connected with route errors (non bloquant)

Ping du serveur Ubuntu depuis le client Windows — réponse ok, le VPN fonctionne :



Ping serveur Ubuntu depuis client Windows — 192.168.2.211 — réponse ok

---

7. Connexion VPN — client Debian

On accède au serveur de partage via curl et on liste les fichiers disponibles :

```
root@staff-netsecure-12:~# curl http://192.168.2.211:8080
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/htm
rict.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<title>Directory listing for /</title>
</head>
<body>
<h1>Directory listing for /</h1>
<hr>
<ul>
<li><a href="ca.crt">ca.crt</a></li>
<li><a href="client/">client/</a></li>
<li><a href="client1-final.ovpn">client1-final.ovpn</a></li>
<li><a href="client1.ovpn">client1.ovpn</a></li>
<li><a href="client2-final.ovpn">client2-final.ovpn</a></li>
<li><a href="dh.pem">dh.pem</a></li>
<li><a href="easy-rsa/">easy-rsa/</a></li>
<li><a href="server/">server/</a></li>
<li><a href="server.conf">server.conf</a></li>
<li><a href="server.crt">server.crt</a></li>
<li><a href="server.key">server.key</a></li>
<li><a href="ta.key">ta.key</a></li>
<li><a href="update-resolv-conf">update-resolv-conf</a></li>
</ul>
<hr>
</body>
</html>
```

curl depuis Debian vers serveur de partage — listing des fichiers .ovpn

On télécharge le fichier client2-final.ovpn :

```
root@staff-netsecure-12:~# curl http://192.168.2.211:8080/client2-final.ovpn -o ~/client2.ovpn
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  8328  100  8328    0     0   877k      0 --:--:-- --:--:-- --:--:--  903k
```

Téléchargement du fichier client2-final.ovpn via curl

On installe OpenVPN et on lance la connexion :

```
apt install -y openvpn
openvpn --config ~/client2.ovpn
```

```
root@staff-netsecure-12:~# openvpn --config ~/client2.ovpn
```

Lancement de la connexion VPN depuis Debian — openvpn --config ~/client2.ovpn

VERIFY OK — connexion établie, IP attribuée 10.8.0.6 :

```
2026-05-11 17:29:27 VERIFY OK: depth=1, CN=VPN-NetSecure
2026-05-11 17:29:27 VERIFY OK: depth=0, CN=server
2026-05-11 17:29:27 Control Channel: TLSv1.3, cipher TLSv1.3 TLS_AES_256_GCM_SHA384, peer certificate: 2048 bit RSA, signat
ure: RSA-SHA256
2026-05-11 17:29:27 [server] Peer Connection Initiated with [AF_INET]192.168.2.211:1194
2026-05-11 17:29:27 TLS: move_session: dest=TM_ACTIVE src=TM_INITIAL reinit_src=1
2026-05-11 17:29:27 TLS: tls_multi_process: initial untrusted session promoted to trusted
2026-05-11 17:29:27 PUSH: Received control message: 'PUSH_REPLY,redirect-gateway def1 bypass-dhcp,dhcp-option DNS 8.8.8.8,d
hcp-option DNS 1.1.1.1,route 10.8.0.1,topology net30,ping 10,ping-restart 120,ifconfig 10.8.0.6 10.8.0.5,peer-id 1,cipher A
ES-256-GCM'
```

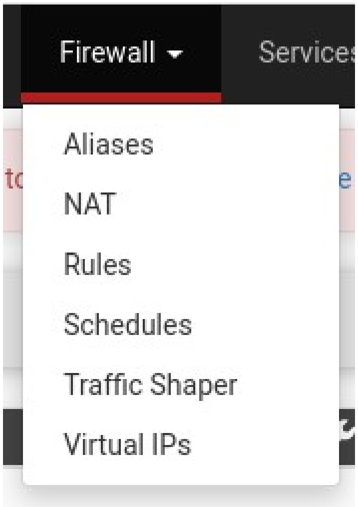
Résultat connexion Debian — VERIFY OK — IP attribuée 10.8.0.6

---

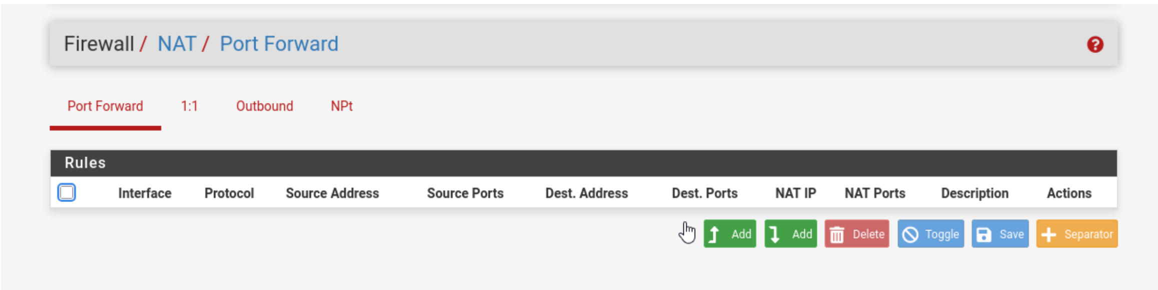
8. Proxy transparent — configuration pfSense

Pour que le proxy soit transparent (les clients ne savent pas qu'ils passent par Squid), on configure une règle NAT sur pfSense qui intercepte le trafic HTTP et le redirige vers Squid :

Firewall > NAT > Port Forward :

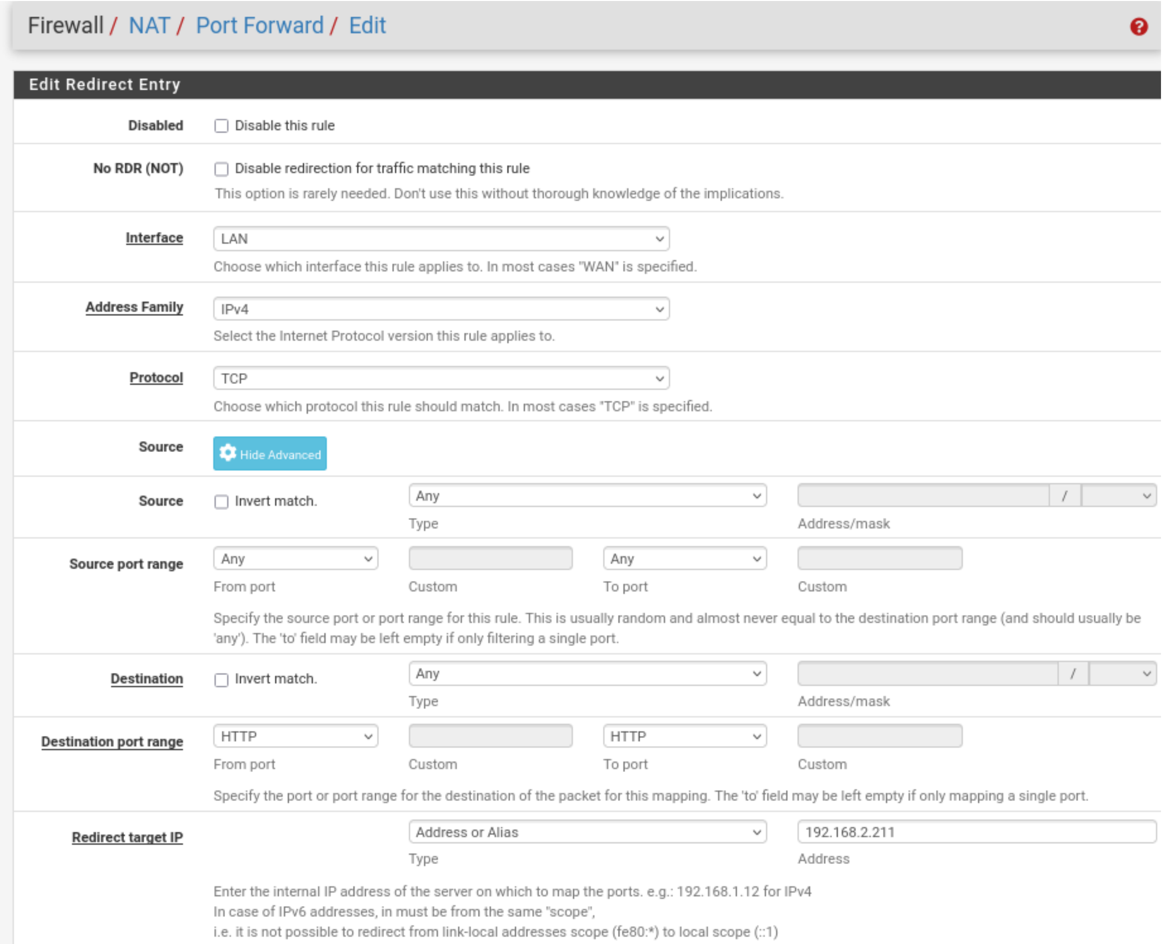


Interface pfSense — menu Firewall > NAT

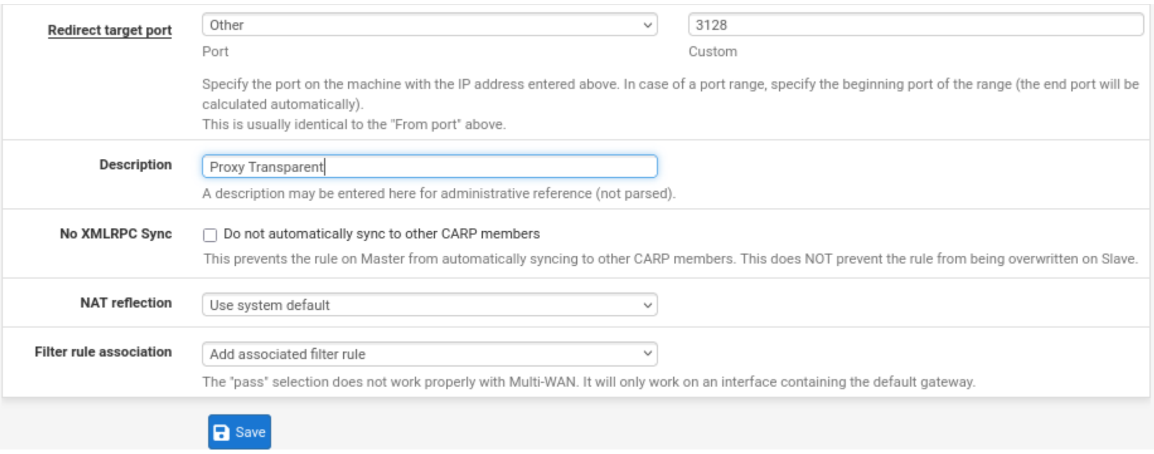


Firewall NAT Port Forward — table vide avant création de la règle

On crée la règle — Interface LAN, protocole TCP, destination port HTTP (80), redirect vers 192.168.2.211:3128 — description Proxy Transparent :



Création de la règle Port Forward — redirect HTTP vers Squid (3128) — description Proxy Transparent



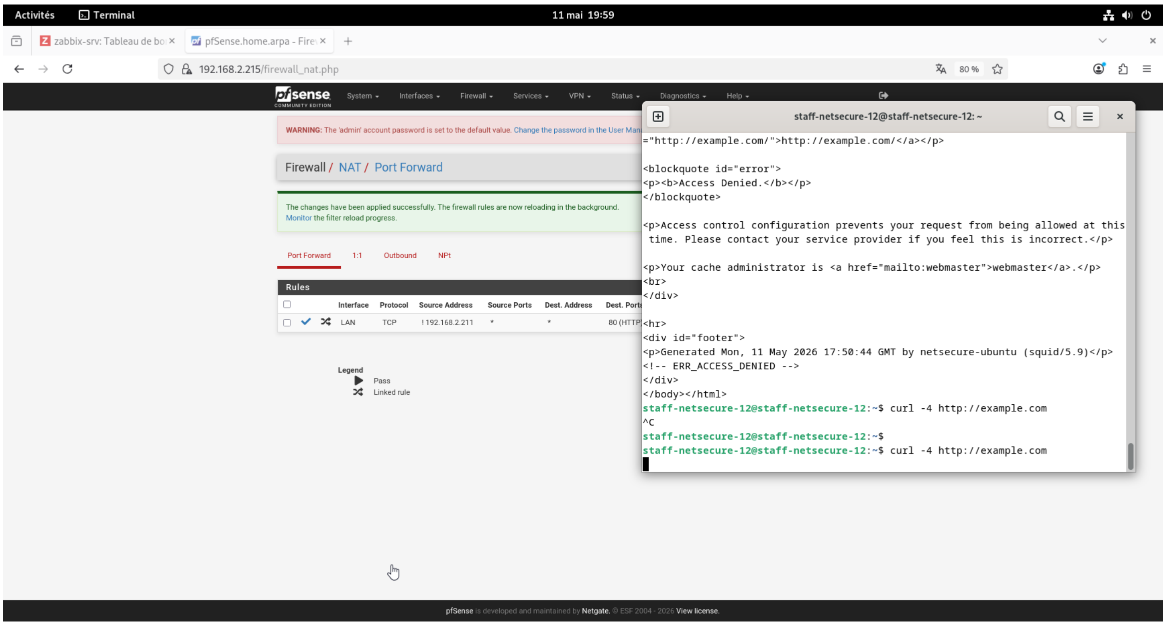
Paramètre Redirect target port — 3128 — description Proxy Transparent — Save

On active le mode intercept dans la config Squid :

Modification squid.conf — nano /etc/squid/squid.conf

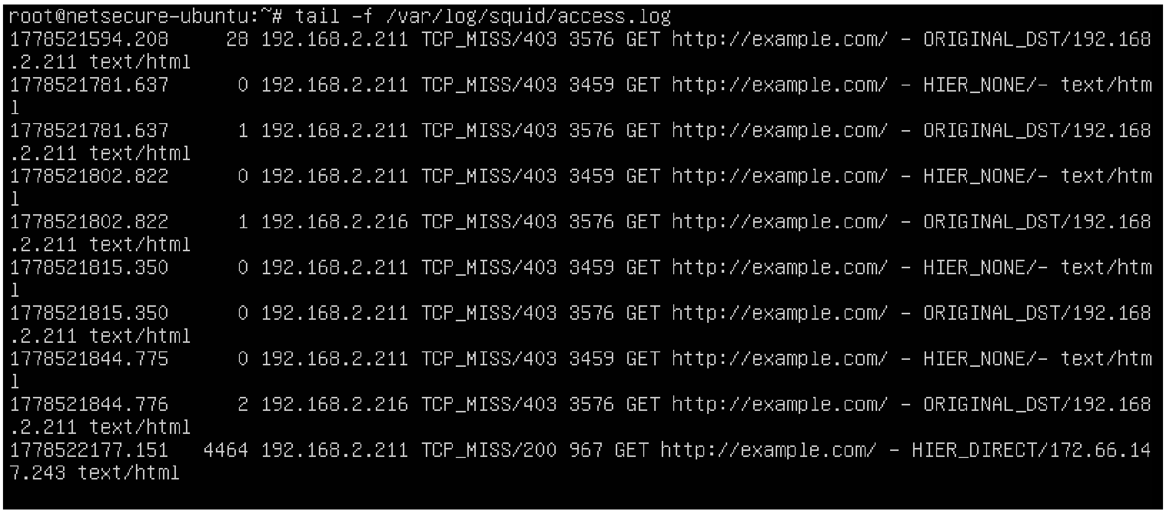
Ajout de la directive intercept — http_port 3128 intercept — mode proxy transparent

Test depuis le client Debian — la page HTTP est bloquée (Access Denied par Squid). La règle pfSense est active :



Résultat final — pfSense Port Forward actif + terminal Debian montrant Access Denied par Squid

Logs Squid côté serveur — TCP_MISS/403 confirme le filtrage transparent en action :



```
root@netsecure-ubuntu:~# tail -f /var/log/squid/access.log
1778521594.208     28 192.168.2.211 TCP_MISS/403 3576 GET http://example.com/ - ORIGINAL_DST/192.168
.2.211 text/html
1778521781.637      0 192.168.2.211 TCP_MISS/403 3459 GET http://example.com/ - HIER_NONE/- text/htm
l
1778521781.637      1 192.168.2.211 TCP_MISS/403 3576 GET http://example.com/ - ORIGINAL_DST/192.168
.2.211 text/html
1778521802.822      0 192.168.2.211 TCP_MISS/403 3459 GET http://example.com/ - HIER_NONE/- text/htm
l
1778521802.822      1 192.168.2.216 TCP_MISS/403 3576 GET http://example.com/ - ORIGINAL_DST/192.168
.2.211 text/html
1778521815.350      0 192.168.2.211 TCP_MISS/403 3459 GET http://example.com/ - HIER_NONE/- text/htm
l
1778521815.350      0 192.168.2.211 TCP_MISS/403 3576 GET http://example.com/ - ORIGINAL_DST/192.168
.2.211 text/html
1778521844.775      0 192.168.2.211 TCP_MISS/403 3459 GET http://example.com/ - HIER_NONE/- text/htm
l
1778521844.776      2 192.168.2.216 TCP_MISS/403 3576 GET http://example.com/ - ORIGINAL_DST/192.168
.2.211 text/html
1778522177.151   4464 192.168.2.211 TCP_MISS/200 967 GET http://example.com/ - HIER_DIRECT/172.66.14
7.243 text/html
```

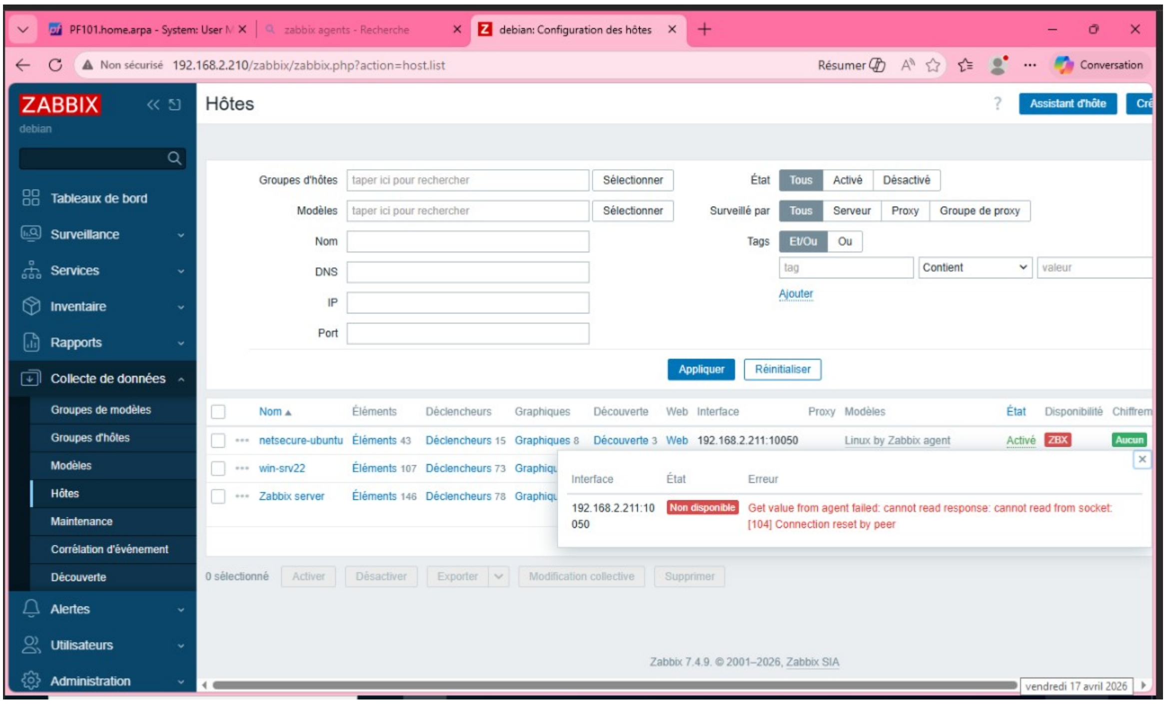
Logs Squid — tail -f /var/log/squid/access.log — TCP_MISS/403 confirme le filtrage

---

9. Annexe — Résolution du problème agent Zabbix Ubuntu

Ces captures concernent le débogage de l'agent Zabbix sur le serveur Ubuntu, effectué dans le cadre du projet NetSecure global. L'agent était actif mais l'interface Zabbix ne le détectait pas.

Interface Zabbix — erreur Cannot read from socket sur netsecure-ubuntu :



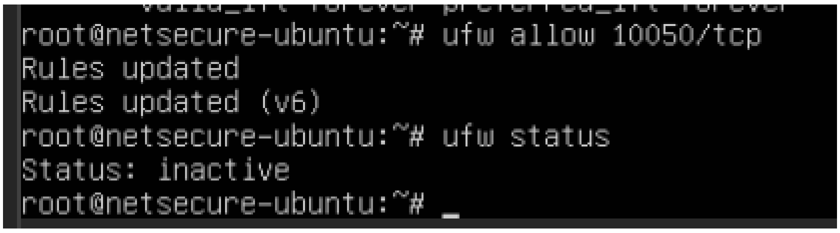
Interface Zabbix — liste des hôtes — erreur agent Ubuntu : cannot read from socket

Vérification du hostname — netsecure-ubuntu :

```
root@netsecure-ubuntu:~# hostname
netsecure-ubuntu
```

Vérification hostname — netsecure-ubuntu

Ouverture du port 10050 via ufw :



```
root@netsecure-ubuntu:~# ufw allow 10050/tcp
Rules updated
Rules updated (v6)
root@netsecure-ubuntu:~# ufw status
Status: inactive
root@netsecure-ubuntu:~# _
```

ufw allow 10050/tcp — règle ajoutée — ufw status : inactive

Vérification de la configuration agent — Server, ServerActive et Hostname correctement configurés :

```
root@netsecure-ubuntu:~# grep -E "^Server|^ServerActive|^Hostname" /etc/zabbix/zabbix_agent2.conf
Server=192.168.2.210
ServerActive=192.168.2.210
Hostname=netsecure-ubuntu
root@netsecure-ubuntu:~# _
```

Vérification config agent — Server=192.168.2.210, ServerActive=192.168.2.210, Hostname=netsecure-ubuntu

ss -tlnp — l'agent écoute bien sur *:10050 — pas de problème d'écoute :

```
root@netsecure-ubuntu:~# ss -tlnp | grep 10050
LISTEN 0      4096          *:10050          *:*     users:(("zabbix_agent2",pid=693,fd=8))
```

*ss -tlnp | grep 10050 — agent écoute bien sur *:10050*

Ping du serveur Zabbix depuis Ubuntu — communication ok :

```
root@netsecure-ubuntu:~# ping 192.168.2.210
PING 192.168.2.210 (192.168.2.210) 56(84) bytes of data.
64 bytes from 192.168.2.210: icmp_seq=1 ttl=64 time=0.635 ms
64 bytes from 192.168.2.210: icmp_seq=2 ttl=64 time=0.462 ms
64 bytes from 192.168.2.210: icmp_seq=3 ttl=64 time=0.497 ms
64 bytes from 192.168.2.210: icmp_seq=4 ttl=64 time=0.478 ms
64 bytes from 192.168.2.210: icmp_seq=5 ttl=64 time=0.422 ms
64 bytes from 192.168.2.210: icmp_seq=6 ttl=64 time=0.563 ms
64 bytes from 192.168.2.210: icmp_seq=7 ttl=64 time=0.479 ms
^C
--- 192.168.2.210 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6120ms
rtt min/avg/max/mdev = 0.422/0.505/0.635/0.065 ms
```

ping 192.168.2.210 depuis Ubuntu — communication ok avec le serveur Zabbix

Consultation des logs de l'agent :

```
root@netsecure-ubuntu:~# tail -50 /var/log/zabbix/zabbix_agent2.log_
```

tail -50 /var/log/zabbix/zabbix_agent2.log — consultation des logs

Logs — Plugin communication protocol version is 6.4.0 — incompatibilité de versions détectée :

```
 on start enabled: false
2026/04/17 11:29:52.771914 using plugin 'NetIf' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: fal
se
2026/04/17 11:29:52.771939 using plugin 'Oracle' (built-in) providing following interfaces: exporter, runner, configurator, maximum capacity: 1000, active check
s on start enabled: false
2026/04/17 11:29:52.771988 using plugin 'PostgreSQL' (/usr/libexec/zabbix/zabbix-agent2-plugin-postgresql) providing following interfaces: exporter, runner, con
figurator, maximum capacity: 1000, active checks on start enabled: false
2026/04/17 11:29:52.772017 using plugin 'Proc' (built-in) providing following interfaces: exporter, collector, maximum capacity: 1000, active checks on start en
abled: false
2026/04/17 11:29:52.772044 using plugin 'ProcExporter' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabl
ed: false
2026/04/17 11:29:52.772086 using plugin 'Redis' (built-in) providing following interfaces: exporter, runner, configurator, maximum capacity: 1000, active checks
 on start enabled: false
2026/04/17 11:29:52.772116 using plugin 'Smart' (built-in) providing following interfaces: exporter, configurator, maximum capacity: 1000, active checks on star
t enabled: false
2026/04/17 11:29:52.772158 using plugin 'Sw' (built-in) providing following interfaces: exporter, configurator, maximum capacity: 1000, active checks on start e
nabled: false
2026/04/17 11:29:52.772184 using plugin 'Swap' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: fals
e
2026/04/17 11:29:52.772210 using plugin 'SystemRun' (built-in) providing following interfaces: exporter, configurator, maximum capacity: 1000, active checks on
start enabled: false
2026/04/17 11:29:52.772248 using plugin 'Systemd' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: f
alse
2026/04/17 11:29:52.772292 using plugin 'TCP' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: false
2026/04/17 11:29:52.772321 using plugin 'UDP' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: false
2026/04/17 11:29:52.772346 using plugin 'Uname' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: fal
se
2026/04/17 11:29:52.772371 using plugin 'Uptime' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: fa
lse
2026/04/17 11:29:52.772412 using plugin 'Users' (built-in) providing following interfaces: exporter, configurator, maximum capacity: 1000, active checks on star
t enabled: false
2026/04/17 11:29:52.772439 using plugin 'VFSDev' (built-in) providing following interfaces: exporter, collector, maximum capacity: 1000, active checks on start
enabled: false
2026/04/17 11:29:52.772479 using plugin 'VFSDir' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: fa
lse
2026/04/17 11:29:52.772509 using plugin 'VfsFs' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: fal
se
2026/04/17 11:29:52.772543 using plugin 'WebCertificate' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start ena
bled: false
2026/04/17 11:29:52.772568 using plugin 'WebPage' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enabled: f
alse
2026/04/17 11:29:52.772592 using plugin 'ZabbixAsync' (built-in) providing following interfaces: exporter, maximum capacity: 1000, active checks on start enable
d: false
2026/04/17 11:29:52.772619 using plugin 'ZabbixStats' (built-in) providing following interfaces: exporter, configurator, maximum capacity: 1000, active checks o
n start enabled: false
2026/04/17 11:29:52.772689 using plugin 'ZabbixSync' (built-in) providing following interfaces: exporter, maximum capacity: 1, active checks on start enabled: f
alse
2026/04/17 11:29:52.774753 Plugin communication protocol version is 6.4.0
2026/04/17 11:29:52.774808 Zabbix Agent2 hostname: [netsecure-ubuntu]
```

Logs agent — Plugin communication protocol version is 6.4.0 — incompatibilité détectée

Version du serveur Zabbix : 7.4.9 — Version de l'agent Ubuntu : 7.4.8 avec protocol 6.4.0 — incompatibilité confirmée. On réinstalle la bonne version de l'agent.

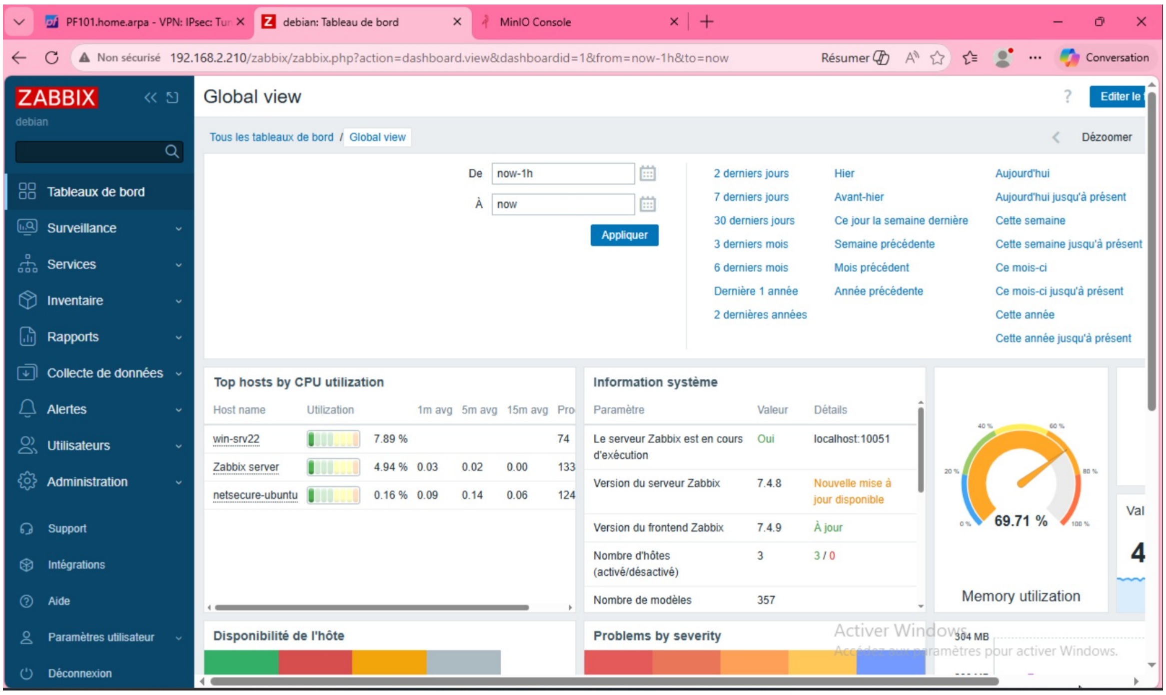
```
root@debian:~# zabbix_server -V
zabbix_server (Zabbix) 7.4.9
Revision 75bd32d3d61 8 April 2026, compilation time: Apr  8 2026 06:11:14

Copyright (C) 2026 Zabbix SIA
License AGPLv3: GNU Affero General Public License version 3 <https://www.gnu.org/licenses/>.
This is free software: you are free to change and redistribute it according to
the license. There is NO WARRANTY, to the extent permitted by law.

This product includes software developed by the OpenSSL Project
for use in the OpenSSL Toolkit (http://www.openssl.org/).

Compiled with OpenSSL 3.0.17 1 Jul 2025
Running with OpenSSL 3.0.19 27 Jan 2026
```

Version serveur Zabbix — 7.4.9

```
root@netsecure-ubuntu:~# zabbix_agent2 -V
zabbix_agent2 (Zabbix) 7.4.8
Revision 6b0f6b25da0 13 March 2026, compilation time: Mar 13 2026 09:38:02, built with: go1.24.10
Plugin communication protocol version is 6.4.0

Copyright (C) 2026 Zabbix SIA
License AGPLv3: GNU Affero General Public License version 3 <https://www.gnu.org/licenses/>.
This is free software: you are free to change and redistribute it according to
the license. There is NO WARRANTY, to the extent permitted by law.

This product includes software developed by the OpenSSL Project
for use in the OpenSSL Toolkit (http://www.openssl.org/).

Compiled with OpenSSL 3.0.2 15 Mar 2022
Running with OpenSSL 3.0.2 15 Mar 2022

We use the library Eclipse Paho (eclipse/paho.mqtt.golang), which is
distributed under the terms of the Eclipse Distribution License 1.0 (The 3-Clause BSD License)
available at https://www.eclipse.org/org/documents/edl-v10.php

We use the library go-modbus (goburrow/modbus), which is
distributed under the terms of the 3-Clause BSD License
available at https://github.com/goburrow/modbus/blob/master/LICENSE
```

Version agent Ubuntu — 7.4.8 — Plugin communication protocol version is 6.4.0 — incompatibilité confirmée

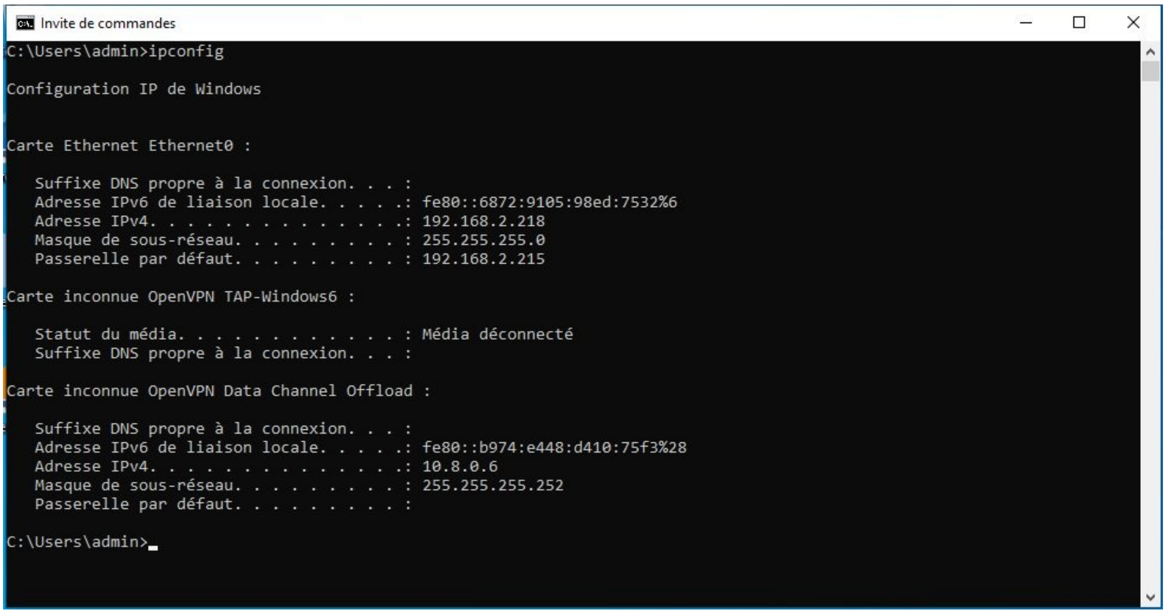
Dashboard Zabbix après résolution — 3 hôtes actifs :



Dashboard Zabbix après résolution — 3 hôtes actifs — win-srv22, Zabbix server, netsecure-ubuntu

10. Résultat final — preuve de connexion VPN

ipconfig sur le client Windows — on voit deux interfaces : Ethernet0 avec l'IP normale 192.168.2.218, et OpenVPN Data Channel Offload avec l'IP VPN 10.8.0.6. La connexion VPN est établie et fonctionnelle :



ipconfig client Windows — IP normale 192.168.2.218 et IP VPN 10.8.0.6 sur OpenVPN Data Channel Offload